

LCA of the Impossible - Myth & Legends

THE TOWER OF BABEL

RAISING A CITY TO HEAVEN,
ONE BRICK AT A TIME.



STEP ONE

Subject & functional unit

SUBJECT

The Tower of Babel is the monument built by a united humanity to make a name and reach the heavens. For engineering purposes, it is reconstructed as a completed form of Etemenanki, Babylon's great seven-stage ziggurat.

FUNCTIONAL UNIT

Provision and operation of one completed 90 m Tower of Babel for one year as a ceremonial stairway and monument, including full construction, one year of maintenance and auxiliary fires.

MODELLING CONVENTION

One tower. One year.

The entire construction burden is charged to the main FU.

A 100-year sensitivity then spreads construction across a plausible service life. The tower may be ancient; the allocation problem is timeless.

STEP ONE

System boundaries & exclusions

INCLUDED

- Sun-dried mudbrick core and regular replastering
- Fired-brick facing and bitumen mortar
- Cedar and other timber, metal fittings and consumed tools
- Material transport to Babylon: direct + WTT without double counting
- Construction fires, one year of repairs and ceremonial fires

EXCLUDED OR SEPARATELY DISCLOSED

- Human labour, food and animal metabolism: no robust allocation
- Language confusion and divine intervention: no measurable physical flow
- Process water emissions: river/canal supply is assumed gravity/manual
- Biogenic CO₂ from wood combustion: disclosed outside the main total
- Active end of life: abandonment and passive erosion are assumed

THE SUPERNATURAL RULE

Magic is not a fuel.

No emissions are assigned unless a physical flow can be reconstructed.

STEP TWO

Life Cycle Inventory (LCI)

ENGINEERING RECONSTRUCTION

The physical analogue is Etemenanki, the ziggurat of Marduk at Babylon and the monument most often connected with the biblical story. Excavation confirms a base close to 91 m square; the Esagil-tablet tradition describes a seven-floor tower within a 90 m cube.

ADOPTED SCALE

- Square base: 90 x 90 m
- Total height: 90 m
- Seven stages including the summit sanctuary
- Mudbrick core with fired-brick facing
- Three monumental approach stairs represented as one masonry volume

FACT VS RECONSTRUCTION

The footprint is attested. The elevation is not.

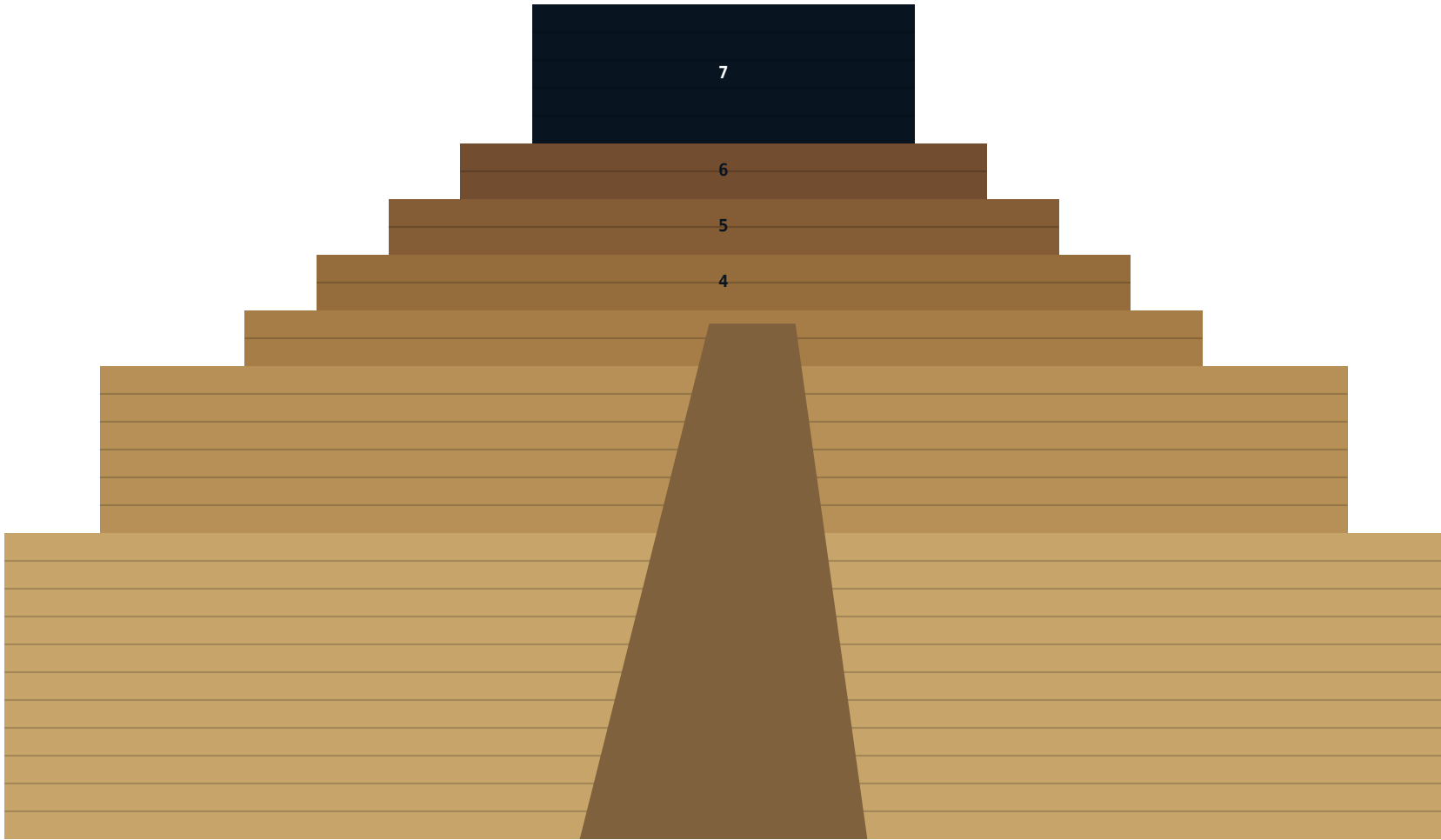
Upper tiers, shell thickness and stair volume remain transparent engineering assumptions.

STEP TWO

Life Cycle Inventory (LCI)

SEVEN LEVELS, SIMPLIFIED

The model converts the idealised tower into seven auditable cuboids. The first stage carries most of the mass; the upper stages retreat toward a cedar-roofed sanctuary.



Stage heights: $33 + 18 + 6 + 6 + 6 + 6 + 15 = 90\text{ m}$

TOTAL MASONRY VOLUME

448,728 m3

VOLUME IN MONUMENTAL STAIRS

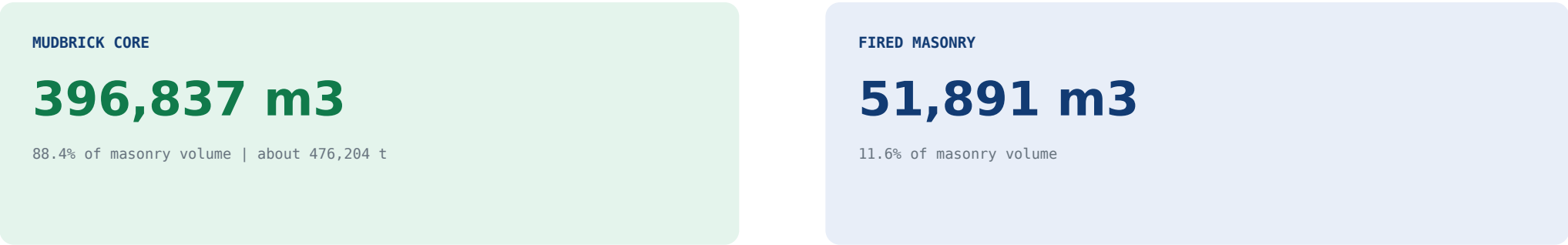
15,000 m3

STEP TWO

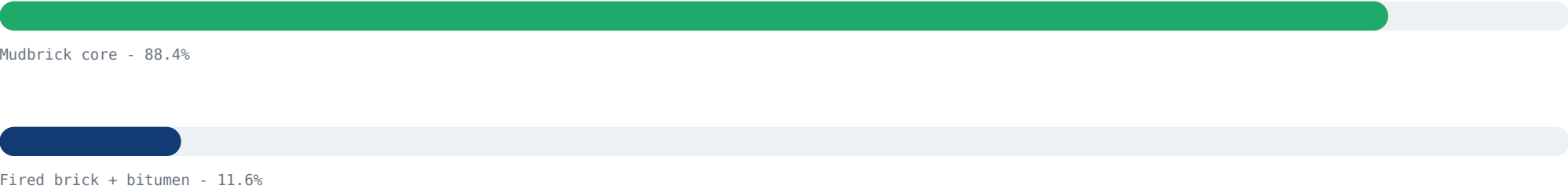
Life Cycle Inventory (LCI)

THE MATERIAL MOUNTAIN

Mesopotamian mudbrick is earth, chopped straw and water, moulded and dried by the sun. The fired shell protects the vulnerable core and carries the tower's darkest carbon secret.



VOLUME SPLIT



The facing is modelled as 90% fired brick and 10% bitumen mortar by volume. Timber and metals sit outside the masonry reconciliation; process water is a temporary input.

STEP TWO

Life Cycle Inventory (LCI)

WHEN 90 METRES BECOME ARITHMETIC

- Mudbrick volume: 396,837 m3
- Manual adobe factor: 0.1143 kg CO2e/m3
- Fired brick mass: 88,734 t
- DEFRA fired-brick proxy: 241.803 kg CO2e/t

SUN-DRIED CORE

45 t CO2e

Large mass, almost no kiln energy

FIRED-BRICK SKIN

21,456 t CO2e

95.4% of the complete one-year result

The adobe factor comes from manual, on-site production and is highly context-sensitive. A modern industrial adobe dataset would add roughly 38,900 t CO2e. The low main value is a deliberate ancient-production analogue, not a universal claim for earth construction.

STEP TWO

Life Cycle Inventory (LCI)

MORTAR, ROOF AND TOOLS

Natural bitumen mortar	5,345 t	210 t CO2e
Cedar and other timber	1,500 t	404 t CO2e
Metal fittings + tools	50 t	191 t CO2e
Process water	119,051 m3	0 t CO2e modelled

Bitumen is represented by DEFRA asphalt, cedar by generic wood and all metals by an average metals factor. These are modern proxies with low historical representativeness - visible in the workbook, not hidden in the legend.

STEP TWO

Life Cycle Inventory (LCI)

SUPPLY TO BABYLON

- Fired bricks: 443,668 tonne.km by road proxy - 56.4 t CO2e
- Bitumen: 1,068,955 tonne.km by small cargo proxy - 18.5 t CO2e
- Timber: 1,200,000 tonne.km by small cargo proxy - 20.7 t CO2e
- Metals: 5,000 tonne.km by road proxy - 0.6 t CO2e

CONSTRUCTION TRANSPORT - DIRECT + WTT

96 t CO2e

Ancient routes, modern transport factors: low representativeness

CONSTRUCTION FIRES

Five hundred tonnes of wood logs represent bitumen heating, food preparation and site fires. The non-biogenic direct + WTT burden enters the main result.

MAIN TOTAL

50 t CO2e

BIOGENIC CO2

718 t CO2

Outside scopes - disclosed separately

STEP TWO

Life Cycle Inventory (LCI)

ONE YEAR ABOVE THE PLAIN

- Mudbrick replastering: 992 m3/year
- Fired-brick replacement: 88.7 t/year
- Bitumen resealing: 53.4 t/year
- Timber replacement: 30 t/year
- Metal replacement: 0.5 t/year

ANNUAL MAINTENANCE

35.9 t CO2e

ANNUAL FIRES

2.0 t CO2e

+ 28.7 t biogenic CO2

THE DAY THE WORDS BROKE

Language confusion changes coordination, not thermodynamics. No fuel, material or measurable energy flow can be attributed to the event. Divine intervention therefore remains outside the inventory.

SUPERNATURAL FLOW

0 kg CO2e

The workers scatter. The bricks remain.

STEP THREE

Results

Global Warming Potential - DEFRA 2026 + literature/engineering proxies

Component / activity	Climate change (kg CO2e)
Mudbrick core	45,353
Fired-brick facing	21,456,059
Bitumen mortar	209,582
Timber	404,256
Metal fittings and tools	191,097
Construction transport	96,210
Construction fires	50,439
Annual maintenance	35,857
Annual operation	2,018

TOTAL

22,491 t CO2e

FIRED-BRICK HOTSPOT

95.4%

DEFRA-derived share of main result: 99.8%

Biogenic CO2 disclosed separately: 746,840 kg CO2

STEP FOUR

Interpretation & conclusion

**The Tower of Babel is not defined by height.
It is defined by fire.**

The sun-dried core holds almost nine tenths of the masonry volume, yet the fired-brick skin carries 95.4% of the complete one-year footprint. The monument's climate story is not the clay itself, but the decision to harden it.

CONSTRUCTION CHARGED TO ONE YEAR

22,491 t CO₂e

CONSTRUCTION SPREAD OVER 100 YEARS

262 t CO₂e/year

98.8% lower than the main FU

A high-carbon kiln factor raises the result to about 123,495 t CO₂e. That range is the episode's largest limitation - and its clearest lesson.

They tried to give one language to the sky.

The kilns answered first.

The words were scattered.

The carbon stayed.

MYTH & LEGENDS

**Where impossible stories meet
measurable emissions**